Test Series No.

1.3

MASTERY TEST SERIES

DURATION: 90 Minutes | **DROPPER** | **M. MARKS: 400**

DATE: 10/02/2026

SYLLABUS

Physics:	Mechanical Properties of Solids, Mechanical Properties of Fluids, Thermal Properties of Matter, Kinetic Theory, Thermodynamics, Oscillations, Waves + 30% Questions from cumulative syllabus of previously conducted part test
Chemistry:	Organic Chemistry - IUPAC Nomenclature, Organic Chemistry – Isomerism, Organic Chemistry - General Organic Chemistry, Hydrocarbons, Purification and Analysis of Organic Compound + 30% Questions from cumulative syllabus of previously conducted part test
Botany:	Respiration in Plants, Photosynthesis in Higher Plants + 30% Questions from cumulative syllabus of previously conducted part test
Zoology:	Locomotion and Movement, Neural Control and Coordination + 30% Questions from cumulative syllabus of previously conducted part test

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **90 minutes** duration and the Test Booklet contains **100** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 400.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Time period of a simple pendulum in a stationary lift is *T*. If the lift accelerates with $\frac{g}{6}$ vertically upwards then the time period will be: (where $g = \text{acceleration due to gravity}$)

(1) $\sqrt{\frac{6}{5}}T$ (2) $\sqrt{\frac{5}{6}}T$

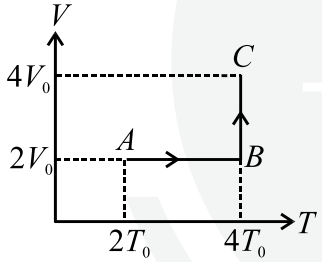
(3) $\sqrt{\frac{6}{7}}T$ (4) $\sqrt{\frac{7}{6}}T$

2. The displacement of a longitudinal wave is represented by the function $s = 10\sin(\omega t - kx)$ cm. If the wavelength of this wave is measured to be 5π cm, find the ratio of the maximum velocity of the particles to the velocity of the wave.

(1) 2 (2) 4

(3) 6 (4) 8

3. Work done by 2 moles of an ideal gas in process ($A \to B \to C$) shown below is, where *V* is volume and *T* is temperature. (*R* is universal gas constant)



(1) $4RT_0$ (2) $4RT_0 ln(2)$

(3) $8RT_0$ (4) $8RT_0 ln(2)$

4. Modulus of Rigidity (in N/m^2) for an ideal gas is:

(1) Zero (2) One

(3) Infinite (4) 0.5

5. A travelling wave is represented by the equation $y = \frac{1}{10}\sin(60t + 2x)$, where *x* and *y* are in meter and *t* is in seconds. For this wave choose the option containing **correct** statements.

A. Wave velocity is 30 m s^{-1}.

B. Frequency of the wave is $\frac{30}{\pi}$ Hz.

C. Wavelength of wave is π m.

D. Particle amplitude is 10 cm.

E. The wave is propagating in +*x* direction.

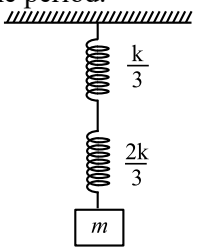
(1) A, B and D only

(2) C, D and E only

(3) A, B, C and D only

(4) A, B, C, D and E

6. A block of mass *m* is suspended using two massless springs of spring constant $\frac{k}{3}$ and $\frac{2k}{3}$ respectively as shown in figure. If the block is slightly displaced vertically downward and released, then it executes SHM with time period.



(1) $2\pi\sqrt{\frac{m}{k}}$ (2) $2\pi\sqrt{\frac{m}{2k}}$

(3) $6\pi\sqrt{\frac{m}{2k}}$ (4) $6\pi\sqrt{\frac{m}{k}}$

7. A block of mass 0.1 kg is held against a wall applying a horizontal force of 5 N on the block. If the coefficient of friction between the block and the wall is 0.5, the magnitude of the frictional force acting on the block is (take $g = 9.8\ \text{m/s}^2$)

(1) 2.5 N (2) 0.98 N

(3) 4.9 N (4) 0.49 N

8. A small block slides without friction down an inclined plane starting from rest. If the distance travelled by the block during the 5th second is s_5 and the distance travelled during the 6th second is s_6, then the ratio s_5 / s_6 is:

(1) 5 / 6 (2) 11 / 9

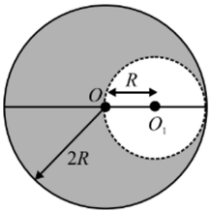
(3) 9 / 11 (4) 7 / 9

9. A machine delivering a constant power of 4.5 W moves a body of mass 1 kg along a straight line. If the body starts from rest, determine the distance moved by the body in 4 seconds.

(1) 8 m (2) 12 m

(3) 16 m (4) 24 m

10. A uniform circular disc of radius R is removed from a bigger circular disc of radius 2R as shown in the figure. The distance of center of mass of the remaining portion is αR from the center of the bigger disc. The value of α is:



(1) 1/3 (2) 2/3

(3) 3/4 (4) 3/5

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11. The angular position of a particle moving in a circle is described by the equation $\theta(t)=\frac{t^3}{3}-2t^2$ radians. Consider the following two statements regarding its motion:

Statement I: The particle reverses its direction of rotation at time $t=4$ seconds.

Statement II: At the exact moment the particle reverses its direction, its angular acceleration is zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct and Statement II is incorrect.
(4) Statement I is incorrect and Statement II is correct.

12. Three planets A, B and C of radius R, 1.5 R and 2R have densities $\rho,\frac{\rho}{2},\frac{\rho}{4}$ respectively. The ratio of acceleration due to gravity ($g_A:g_B:g_C$) at the surface of these planets is:

(1) 3 : 4 : 2
(2) 2 : 4 : 3
(3) 4 : 2 : 3
(4) 4 : 3 : 2

13. A nurse is setting up an IV drip for a patient. The gauge pressure inside the patient's vein is 2000 Pa. The density of the saline solution being administered is $1000\ \text{kg/m}^3$ and take $g=10\ \text{m/s}^2$.

Statement I: To ensure the fluid enters the vein, the minimum height of the IV bag above the needle must be 0.2 meters.

Statement II: If the saline solution is replaced by a denser fluid while keeping the bag at the same height, the fluid will fail to enter the vein.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct and Statement II is incorrect.
(4) Statement I is incorrect and Statement II is correct.

14. A student is performing an experiment to find the diameter of a metal cylinder. After taking the measurement, he records the value in his lab notebook as 5.20 cm. Based on the accuracy of this reported value, which of the following instruments was most likely used for the measurement?

(1) A regular meter scale where the smallest marking is 1 mm.
(2) A vernier calliper where the main scale is divided into millimetres and 10 divisions on the vernier scale match 9 divisions on the main scale.
(3) A screw gauge with a pitch of 1 mm and 100 divisions on the circular scale.
(4) A screw gauge with a pitch of 0.5 mm and 50 divisions on the circular scale.

15. Bulk modulus of a spherical object is $2\times10^{11}\ \text{N/m}^2$. The change in pressure required to increase density of material by 0.3% is:

(1) $6\times10^8\ \text{N/m}^2$ (2) $6\times10^9\ \text{N/m}^2$
(3) $8\times10^8\ \text{N/m}^2$ (4) $8\times10^9\ \text{N/m}^2$

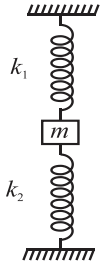
16. An uncalibrated thermometer reads –10° at the freezing point of water and 190° at the boiling point. What is the true temperature in Celsius when this thermometer reads 50°?

(1) 25°C (2) 30°C
(3) 40°C (4) 60°C

17. The most probable speed of the molecules in a sample of an ideal gas is measured to be 200 m/s. What is the root-mean-square (rms) speed of the molecules in this same gas sample?

(1) $200\sqrt{\frac{2}{3}}$ m/s (2) 300 m/s
(3) $100\sqrt{6}$ m/s (4) $200\sqrt{3}$ m/s

18. A block of mass $m=4\ \text{kg}$ is suspended between two springs with spring constants $k_1=100\ \text{N/m}$ and $k_2=300\ \text{N/m}$. If the block is displaced vertically and released, what is the frequency of its simple harmonic motion?



(1) $\frac{5}{\pi}$ Hz (2) $\frac{10}{\pi}$ Hz
(3) $\frac{20}{\pi}$ Hz (4) $\frac{40}{\pi}$ Hz

[Contd...

19. The equation of simple harmonic motion of a block of mass 1 kg along x-axis with mean position at the origin is given by $x = 2\sin(4\pi t + \frac{\pi}{6})$ m where t is time in second. Match the quantities asked in List-I with their values in List-II.

List-I		List-II	
(A)	Maximum speed of block (in m/s)	(I)	0.5
(B)	Maximum acceleration of block (in m/s^2)	(II)	$16\pi^2$
(C)	Force constant (in N/m)	(III)	$32\pi^2$
(D)	Time period of block (in second)	(IV)	8π

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-II, D-I
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-III, B-II, C-I, D-IV

20. The breaking stress of steel is 4.5×10^8 N/m^2. The minimum diameter of a steel cable required to support a load of 900 kg is: (g = 10 m/s^2)

(1) 0.5 cm
(2) 1 cm
(3) 0.4 cm
(4) 0.8 cm

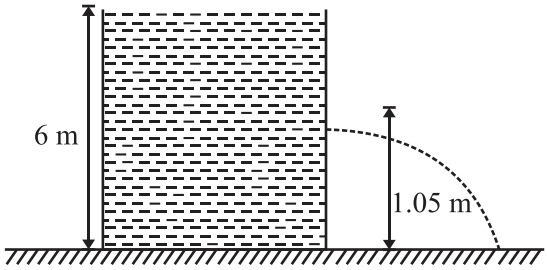
21. A thin copper wire of length L increases in length by 1% when heated from 0°C to 100°C. If a thin square copper plate of area $L \times L$ is heated from 0°C to 100°C, the percentage increase in its area will be:

(1) 1%
(2) 2%
(3) 3%
(4) 4%

22. In a closed organ pipe of length l, the fundamental frequency of sound wave is: (v = speed of sound)

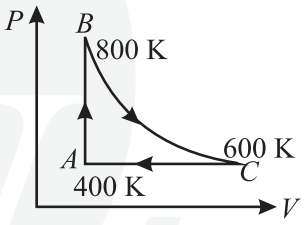
(1) $v / 2l$
(2) v / l
(3) $v / 4l$
(4) $3v / 4l$

23. Water is filled in a cylindrical container to a height of 6 m. An orifice is made at a height of 1.05 m from the base of the container. The ratio of the cross-sectional area of the orifice to that of the container is 0.1. Determine the square of the speed of the liquid coming out from the orifice. Use $g = 10$ m/s^2.



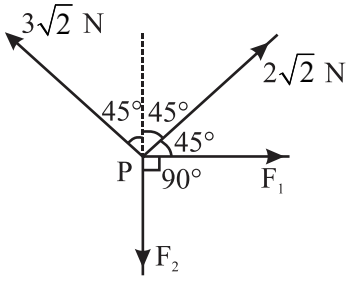
(1) 80 m^2/s^2 (2) 100 m^2/s^2
(3) 120 m^2/s^2 (4) 150 m^2/s^2

24. One mole of diatomic ideal gas undergoes a cyclic process $ABCA$ as shown in figure. The process BC is adiabatic. The temperatures at A, B and C are 400 K, 800 K and 600 K, respectively. Choose the **correct** statement.



(1) The change in internal energy in whole cyclic process is 250 R
(2) The change in internal energy in the process CA is 700 R
(3) The change in internal energy in the process AB is -350 R
(4) The change in internal energy in the process BC is -500 R

25. Four forces are acting at a point P in equilibrium as shown in figure. The ratio of force F_1 to F_2 is 3 : x. Find x.



(1) 5 (2) 10
(3) 15 (4) 20

26. Arrange the following carbanions in decreasing order of stability:

(A) $\overset{\ominus}{C}H_2 - Cl$ (B) $\overset{\ominus}{C}H_2 - CN$

(C) $\overset{\ominus}{C}H_2 - NO_2$ (D) $\overset{\ominus}{C}H_2 - CH_3$

(1) C > B > D > A
(2) C > B > A > D
(3) C > A > B > D
(4) B > C > A > D

27. The ratio of angular momentum of electron in two successive orbit is a (a > 1) and their difference is b. Then a/b is equal to:

(1) $\frac{n}{n+1}$
(2) $\frac{n+1}{n}$
(3) $\frac{n+1}{n}.\frac{h}{2\pi}$
(4) $\frac{n+1}{n}.\frac{2\pi}{h}$

28. In the given statements:

A. Acetylene is colourless, odourless gas.
B. Acetylene is more acidic than Propyne.
C. Baeyer's reagent is acidified $KMnO_4$ solution.
D. But-2-yne on ozonolysis forms only acetic acid.
E. Bromine water can be used to distinguish ethyne and ethene.

Choose the **correct** statement:
(1) A, B, C & D only
(2) B, D only
(3) B, C & D only
(4) B, D & E only

29. Given below are two statements.

Statement I: In partition chromatography, stationary phase is thin film of liquid present in the inert support.

Statement II: In paper chromatography, the material of paper acts as a stationary phase.

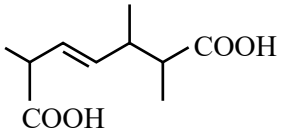
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct and Statement II is incorrect.
(2) Statement-I is incorrect and Statement-II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

30. Given below are two statements.

Statement I: The electronegativity of group 14 elements from Si to Pb gradually decreases.

Statement II: Group 14 contains non-metallic, metallic, as well as metalloid elements.

In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct and Statement II is incorrect.
(2) Statement-I is incorrect and Statement-II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

31. The IUPAC name of  :

(1) 6-Carboxy-2,3-dimethyl-4-heptenoic acid
(2) 2,3,6-Trimethyl-1-heptene-1,7-dioic acid
(3) 2,5,6-Trimethyl-3-heptenedioic acid
(4) 6-Carboxy-2,5-dimethyl-3-heptenoic acid

32. In following reactions:

$$CH_3CH_2C \equiv CH \xrightarrow[(ii)\ CH_3CH_2Br]{(i)\ NaNH_2} (X) \xrightarrow{H_2/Pd-BaSO_4} (Y) \xrightarrow{\text{Alkaline } KMnO_4} (Z)$$

Name of compound (Z) is:
(1) Hexane-3,4-diol (2) Hexane-1,2-diol
(3) Hexene-2,4-diol (4) Hexane-1,3-diol

33. Consider a hypothetical equilibrium:

$A(aq.) \rightleftharpoons 2B(aq.) + C(aq.)$

Before the reaction, A and B were present such that mole fraction of B is 1/3 as well as molar concentration of B is M. At equilibrium the reaction mixture has mole fraction of A equal to 1/2. The equilibrium constant of the reaction can be given by:

(1) $\frac{27M^3}{28}$ (2) $\frac{28M^3}{27}$
(3) $\frac{M^2}{7}$ (4) $\frac{9M^2}{28}$

34. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The boiling point of cis-1,2-Dichloroethene is higher than that of corresponding trans isomer.

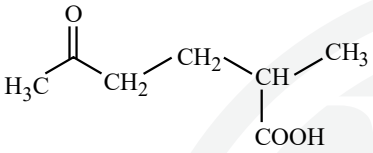
Reason (R): cis-1,2-Dichloroethene has higher dipole moment as compared to that of the trans isomer.

In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both (A) and (R) are true and the (R) is a correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not a correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

35. The technique used for purification of steam volatile water immiscible substance is:
(1) Fractional distillation
(2) Fractional distillation under reduced pressure
(3) Distillation
(4) Steam distillation

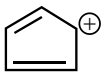
36. Which of the following represents the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar?
(1) Ba < Ca < Se < S < Ar
(2) Ca < Ba < S < Se < Ar
(3) Ca < S < Ba < Se < Ar
(4) S < Se < Ca < Ba < Ar

37. The correct IUPAC nomenclature for the following compound is:



(1) 5-Formyl-2-methylhexanoic acid
(2) 2-Methyl-5-oxohexanoic acid
(3) 2-Formyl-5-methylhexan-6-oic acid
(4) 5-Methyl-2-oxohexan-6-oic acid

38. A hydrocarbon 'X' with formula C_6H_8 uses two moles of H_2 on catalytic hydrogenation of its one mole. On ozonolysis, 'X' yields two moles of methane dicarbaldehyde. The hydrocarbon 'X' is:
(1) hexa-1, 3, 5-triene
(2) 1-methylcyclopenta-1,4-diene
(3) cyclohexa-1, 3-diene
(4) cyclohexa-1, 4-diene

39. Which of the following is an aromatic compound?
(1) 
(2) 
(3) 
(4) 

40. Match **List-I** with **List-II**.

List-I		List-II	
(A)	XeO_3	(I)	Planar triangular
(B)	$XeOF_4$	(II)	T-shape
(C)	BO_3^{3-}	(III)	Trigonal pyramid
(D)	ClF_3	(IV)	Square pyramid

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-I, C-III, D-II

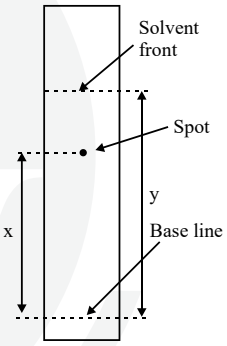
41. How many different alkenes (including stereoisomers) with molar mass 84 upon hydrogenation can give 3-methyl pentane?
(1) 3
(2) 4
(3) 5
(4) 6

42. Compound A formed in the following reaction reacts with B gives the product C. Find out A and B.

$$CH_3-C\equiv CH + Na \rightarrow A \xrightarrow{B} CH_3-C\equiv C-\underset{(C)}{CH_2-CH_2}-\underset{CH_3}{CH} + NaBr$$

(1) $A = CH_3-C\equiv C^-Na^+$, $B = CH_3-CH_2-CH-Br$
(2) $A = CH_3-CH=CH_2$, $B = CH_3-CH_2-CH_2-Br$
(3) $A = CH_3-CH_2-CH_3$, $B = CH_3-C\equiv CH$
(4) $A = CH_3-C\equiv \bar{C}Na^+$, $B = CH_3-CH_2-CH_3$

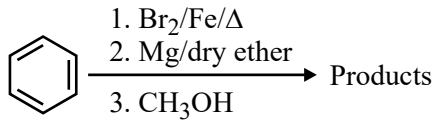
43. In the following developed chromatogram:

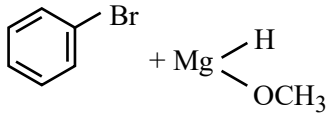


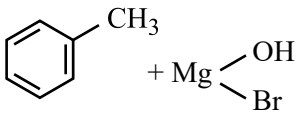
Correct option regarding R_f (retardation factor) is:
(1) $R_f = x$
(2) $R_f = y$
(3) $R_f = \frac{x}{y}$
(4) $R_f = \frac{y}{x}$

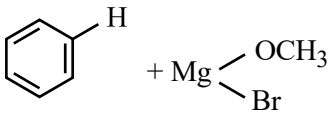
44. In the given statements:
A. Number of pi bonds in Br_3O_8 are 8.
B. Number of electrons lost per molecule of Fe_3O_4 during its oxidation to Fe_2O_3 is one.
C. The difference in oxidation number of Cl atoms in $CaOCl_2$ is zero.
D. The difference in the oxidation number of the two types of sulphur atoms in $Na_2S_4O_6$ is 5.
Choose the correct statements:
(1) A, B & C only
(2) B, C & D only
(3) A, B & D only
(4) A, C & D only

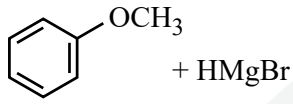
45. For following sequence of reactions, the correct products are:



(1) 

(2) 

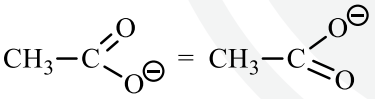
(3) 

(4) 

46. Which of the following resonance structures does **not** represent the **correct** stability order:

(1) $CH_3-CH_2-\overset{\oplus}{C}H-\ddot{O}CH_3 < CH_3-CH_2-CH=\overset{\oplus}{O}CH_3$

(2) $CH_2=CH-\overset{\ominus}{C}H-CH=\overset{\oplus}{N}H_2 > \overset{\ominus}{C}H_2-CH=CH-CH=\overset{\oplus}{N}H_2$

(3) 

(4) 

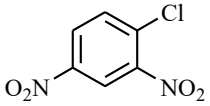
47. Match **List-I** with **List-II**.

List-I		List-II	
(A)	16 g of $CH_4(g)$	(I)	Weighs 28 g
(B)	1 g of $H_2(g)$	(II)	60.2×10^{23} electrons
(C)	1 mole of $N_2(g)$	(III)	Weighs 64 g
(D)	1 mol of $SO_2(g)$	(IV)	Occupies 11.4L volume at STP

Choose the **correct** answer from the options given below:

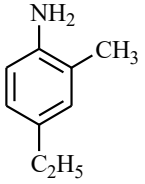
(1) A-I, B-III, C-II, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-II, B-IV, C-III, D-I
(4) A-II, B-IV, C-I, D-III

48. Given below are two statements.

Statement I: 

Compound - A

IUPAC name of compound A is 4-chloro-1,3-dinitrobenzene

Statement II: 

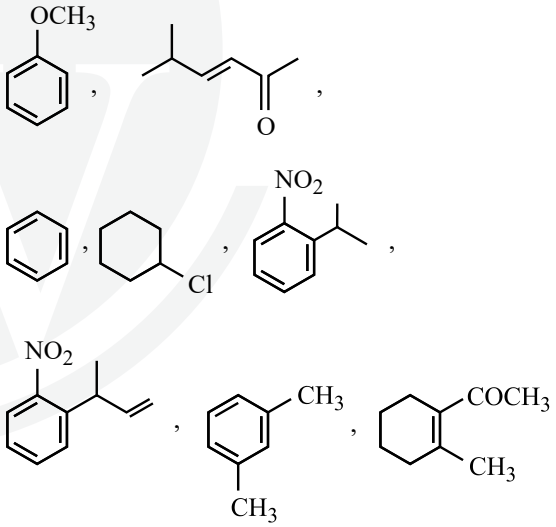
Compound - B

IUPAC name of compound B is 4-ethyl-2-methylaniline.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct and Statement II is incorrect.
(2) Statement-I is incorrect and Statement-II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

49. How many compounds among the following compounds show inductive, mesomeric as well as hyperconjugation effects?



(1) 1 (2) 2
(3) 3 (4) 4

50. He, N_2 and O_3 are expanded adiabatically and their expansion curves between log P and log V are plotted under similar conditions. About the ratio of the slopes, which one is **incorrect**?

(1) The ratio of the slopes of log P vs log V curves for He and O_3 is 1.25
(2) The ratio of slopes of log P vs log V curves for He and N_2 is 1.19.
(3) The ratio of slopes of log P vs log V curves for N_2 and O_3 is 1.05
(4) The slope of He is least steeper and for O_3 is most steeper.

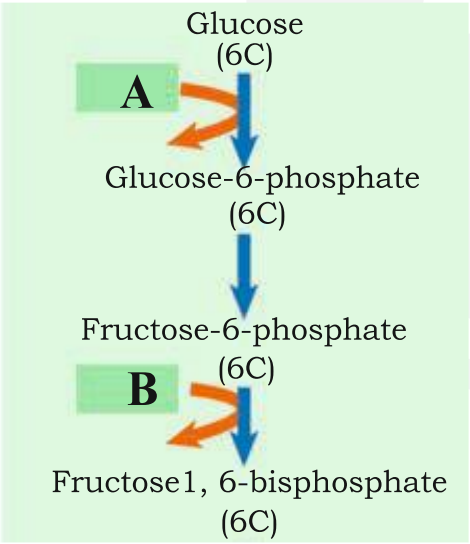
51. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Telophase	(I)	Initiation of condensation of chromosomal material
(B)	Prophase	(II)	Nucleolus, Golgi complex and ER reform
(C)	Metaphase	(III)	Centromeres split and chromatids separate
(D)	Anaphase	(IV)	Spindle fibres attach to the kinetochores of chromosomes

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV
(2) A-III, B-I, C-IV, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-II, D-I

52. Identify the molecules represented by 'A' and 'B' for the given pathway.



(1) A - ATP; B - ATP
(2) A - ATP; B - NADPH
(3) A - NADH; B - ATP
(4) A - ADP; B - ATP

53. Bulliform cells in grasses are modified cells of:

(1) abaxial epidermis.
(2) adaxial epidermis.
(3) vascular bundle sheath.
(4) spongy parenchyma.

54. Given below are two statements.

Statement I: When fats are used in respiration, the RQ is more than 1.

Statement II: When proteins are respiratory substrates, the respiratory ratio would be about 0.9.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct and Statement II is incorrect.
(4) Statement I is incorrect and Statement II is correct.

55. The number of species that are known and described ranges between:

(1) 1.7-1.8 million.
(2) 17-18 million.
(3) 17-18 trillion.
(4) 7-8 billion.

56. In the chloroplast, the enzyme NADP reductase is located:

(1) on the lumen side of the thylakoid membrane.
(2) between the outer and inner membrane.
(3) on the stroma side of the thylakoid membrane.
(4) on the outer membrane.

57. Read the following statements (A-E) regarding respiration in plants.

A. Plants, unlike animals, have no specialised organs for gaseous exchange.
B. Plants have stomata and lenticels.
C. Each plant part takes care of its own gas-exchange needs.
D. There is very little transport of gases from one plant part to another.
E. When cells photosynthesise, the availability of O_2 is a problem in these cells since O_2 is released within the cell.

How many of the above statements are **incorrect**?

(1) Four
(2) One only
(3) Two only
(4) Three only

58. To produce 2 molecules of glucose through the Calvin cycle, how many molecules of CO_2 must be fixed and how many turns of the cycle are required?

(1) 6 CO_2 molecules and 6 turns

(2) 12 CO_2 molecules and 12 turns

(3) 24 CO_2 molecules and 6 turns

(4) 24 CO_2 molecules and 12 turns

59. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): In fermentation there is a net gain of only two molecules of ATP for each molecule of glucose degraded to pyruvic acid.

Reason (R): NADH is oxidised to NAD^+ rather slowly in fermentation as compared to aerobic respiration.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and the (R) is a correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not a correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

60. Polysomes are:

(1) rRNA associated with ribosomes (consisting of subunits 50S and 30S).

(2) mRNA associated with ribosomes (consisting of subunits 40S and 30S).

(3) rRNA associated with ribosomes (consisting of subunits 50S and 20S).

(4) mRNA associated with ribosomes (consisting of subunits 50S and 30S).

61. Saturation of photosynthesis in C_3 plants is observed only beyond:

(1) 360 $\mu l L^{-1}$.

(2) 400 $\mu l L^{-1}$.

(3) 450 $\mu l L^{-1}$.

(4) 330 $\mu l L^{-1}$.

62. During glycolysis the conversion of glucose to glucose-6-phosphate is a/an _____ reaction.

(1) oxidation

(2) reduction

(3) phosphorylation

(4) decarboxylation

63. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Diatoms	(I)	Gelatinous sheath
(B)	*Nostoc*	(II)	The cell wall forms two thin overlapping shells
(C)	Dinoflagellates	(III)	Stiff cellulose plates
(D)	Euglenoids	(IV)	Pellicle

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-III, D-I

(2) A-III, B-I, C-IV, D-II

(3) A-II, B-I, C-III, D-IV

(4) A-II, B-IV, C-I, D-III

64. Complex IV contains two copper centers and:

(1) Cytochrome *b* and c_1.

(2) Cytochrome *a* and a_3.

(3) Cytochrome a_1 and a_2.

(4) Cytochrome a_2 and c_1.

65. Identify the **correctly** matched pair regarding the colour of pigments observed in a paper chromatogram.

(1) Chlorophyll *a* - Yellow green

(2) Chlorophyll *b* - Bright or blue green

(3) Xanthophylls - Yellow

(4) Carotenoids - Blue

66. Glycerol enters the respiratory pathway after being converted into:

(1) acetyl CoA.

(2) pyruvic acid.

(3) PGAL (3-Phosphoglyceraldehyde).

(4) oxaloacetic acid.

67. Identify the **correct** statements regarding red algae.

A. The majority of the red algae are marine.
B. The red thalli of most of the red algae are multicellular.
C. Sexual reproduction is oogamous.
D. Agar is obtained from members of red algae.
E. Carrageen is obtained from red algae.

Choose the **correct** options:

(1) A, B, C and E only
(2) A, B, C and D only
(3) B, C, D and E only
(4) A, B, C, D and E

68. CF_0 is embedded in the thylakoid membrane and carries out:

(1) facilitated diffusion of electrons across the membrane.
(2) facilitated diffusion of protons across the membrane.
(3) active transport of protons across the membrane.
(4) active transport of electrons across the membrane.

69. Identify the **true** statement regarding limiting factors of photosynthesis.

(1) Light is the only limiting factor.
(2) Water is the most important limiting factor.
(3) Carbon dioxide is the major limiting factor.
(4) Temperature does not affect the rate of photosynthesis.

70. In yeast, the anaerobic breakdown of pyruvic acid results in the formation of:

(1) lactic acid and CO_2.
(2) ethanol and CO_2.
(3) acetyl CoA and H_2O.
(4) lactic acid and H_2O.

71. The splitting of water in the chloroplast:

(1) is associated with the PS II.
(2) is associated with the PS I.
(3) occurs in the stroma.
(4) creates carbon dioxide, one of the net products of photosynthesis.

72. In C_4 plant malic acid or aspartic acid is formed in:

(1) bundle sheath cells.
(2) epidermal cells.
(3) mesophyll cells.
(4) xylem cells.

73. Identify the **correct** statement regarding the Krebs' cycle.

(1) During the conversion of succinyl-CoA to succinic acid a molecule of NADH + H^+ is synthesised.
(2) There are only two points in the cycle where NAD^+ is reduced to NADH + H^+.
(3) There are two points in the cycle where FAD^+ is reduced to $FADH_2$.
(4) The TCA cycle requires the continued replenishment of oxaloacetic acid, the first member of the cycle.

74. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): During glycolysis the conversion of BPGA to 3-phosphoglyceric acid (PGA), is an energy yielding process.

Reason (R): Two redox-equivalents are removed from PGAL and transferred to a molecule of NAD^+.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and (R) is a correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not a correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

75. Identify the type of phyllotaxy shown in the image and select the **correct** option.



(1) Alternate (2) Palmate
(3) Opposite (4) Whorled

76. Which of the following is **correct** regarding the types of movements exhibited by human cells?
(1) Macrophages move using ciliary movement to reach the site of infection.
(2) Streaming of protoplasm is the primary mechanism for movement in human limbs.
(3) Microfilaments are cytoskeletal elements involved in the amoeboid movement of leucocytes.
(4) The movement of ova through the female reproductive tract is primarily facilitated by muscular movement.

77. The Somatic Neural System (SNS) is responsible for relaying impulses from;
(1) CNS to involuntary organs and smooth muscles.
(2) involuntary organs to the CNS.
(3) CNS to skeletal muscles.
(4) viscera to the central nervous system.

78. In a myosin monomer, which structural component forms the cross arm?
(1) Light meromyosin forming the tail region.
(2) Heavy meromyosin consisting of a globular head and a short arm.
(3) The regulatory proteins troponin and tropomyosin.
(4) Polymerization of G-actin subunits to form F-actin.

79. Which of the following **correctly** describes the condition of the axonal membrane during the resting state of a neuron?
(1) It is freely permeable to sodium ions (Na^+).
(2) It is nearly impermeable to potassium ions (K^+).
(3) It is impermeable to negatively charged proteins present in the axoplasm.
(4) It maintains a positive charge on the inner surface of the membrane.

80. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Fibrous joint	(I)	Between carpals
(B)	Cartilaginous joint	(II)	Between flat skull bones
(C)	Ball and Socket joint	(III)	Between adjacent vertebrae
(D)	Gliding joint	(IV)	Between humerus and pectoral girdle

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-IV, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-IV, B-I, C-II, D-III
(4) A-II, B-IV, C-I, D-III

81. Which part of the human brain acts as the major coordinating center for sensory and motor signaling?
(1) Hypothalamus
(2) Thalamus
(3) Cerebrum
(4) Corpora quadrigemina

82. Given below are two statements:
Statement I: Basophils are the most abundant cells among total WBCs and are phagocytic in nature.
Statement II: Plasma without clotting factors is called serum.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

83. During the resting state of a muscle fiber, the active binding sites for myosin on the actin filaments are masked by:
(1) a subunit of troponin.
(2) the tail of the light meromyosin.
(3) the globular head of the heavy meromyosin.
(4) two filaments of F-actin helically wound together.

84. Which of the following statements regarding the regulation of kidney function are **correct**?
A. An increase in body fluid volume stimulates osmoreceptors to release ADH.
B. Angiotensin II is a powerful vasodilator that decreases glomerular blood pressure.
C. A fall in GFR activates JG cells to release renin.
D. Atrial Natriuretic Factor acts as a check on the renin-angiotensin mechanism.
(1) Only C and D are correct
(2) A and B are correct
(3) A, C and D are correct
(4) All are correct

85. Identify the **correct** statements regarding the structure of a neuron:

A. Bipolar neurons are found in the retina of the eye.
B. Unipolar neurons are found usually in the embryonic stage.
C. Multipolar neurons have one axon and one dendrite.
D. Myelinated nerve fibers are found in spinal and cranial nerves.

(1) A, C, and D (2) A, B, and D
(3) B, C, and D (4) A, B and C

86. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Myasthenia gravis leads to paralysis of skeletal muscles.

Reason R: It is an autoimmune disorder that affects the neuromuscular junction.

In the light of the above statements, choose the **correct** answer from the options given below:

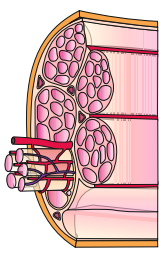
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

87. Consider the following statements regarding the human Cardiac Cycle and choose the **correct** option.

A. The duration of a cardiac cycle is 0.8 seconds.
B. During joint diastole, all four chambers are in a relaxed state and semilunar valves are open.
C. Atrial systole increases the flow of blood into the ventricles by about 30 percent.
D. The second heart sound (dub) is associated with the closure of the tricuspid and bicuspid valves.

(1) Only A and C are correct
(2) A, B and C are correct
(3) B and D are correct
(4) A and B are correct

88. Based on the given figure, what is the **correct** sequence of layers in a muscle tissue from outside to inside?



(1) Muscle fibre → Sarcolemma → Fascicle → Fascia
(2) Fascia → Fascicle → Muscle fibre → Sarcolemma
(3) Sarcolemma → Muscle fibre → Fascicle → Fascia
(4) Fascicle → Fascia → Sarcolemma → Muscle fibre

89. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Hypothalamus	(I)	Controls body temperature and urge for eating
(B)	Cerebellum	(II)	Convoluted surface to provide space for more neurons
(C)	Medulla	(III)	Connects two cerebral hemispheres
(D)	Corpus callosum	(IV)	Controls respiration and gastric secretions

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-IV, D-III
(2) A-I, B-IV, C-II, D-III
(3) A-I, B-II, C-IV, D-III
(4) A-I, B-II, C-III, D-IV

90. Consider the following statements about the sliding filament theory:

A. Contraction is initiated by a signal sent by the CNS via a sensory neuron.
B. During contraction, the 'I' bands get reduced while 'A' bands retain their length.
C. The H-zone is the central part of the thick filament not overlapped by thin filaments.
D. Shortening of the sarcomere occurs when 'Z' lines are pulled inwards.

Which of the statements are **correct**?

(1) A, B, and C only
(2) B, C, and D only
(3) A and D only
(4) All are correct

91. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Transmission of an impulse across an electrical synapse is faster than across a chemical synapse.

Reason R: Electrical synapses use neurotransmitters to bridge the synaptic cleft.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) A is false but R is true.
(3) A is true but R is false.
(4) Both A and R are true but R is not the correct explanation of A.

92. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Fibrinogen	(I)	Osmotic balance
(B)	Globulins	(II)	Involved in defense mechanisms
(C)	Albumins	(III)	Coagulation of blood
(D)	Spleen	(IV)	Graveyard of RBCs

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-I, B-IV, C-III, D-II

93. Which of the following is **true** regarding the neural organization in different animals?

(1) In *Hydra*, the neural system is composed of a complex brain and ganglia.
(2) Insects have a better-organized system than *Hydra*, featuring a brain and ganglia.
(3) Vertebrates have a simple network of neurons without a central processing site.
(4) The endocrine system provides point-to-point connections for quick coordination.

94. White fibers are different from red fibers because white fibers have:

(1) high myoglobin content and more mitochondria.
(2) high amount of sarcoplasmic reticulum and depend on anaerobic process for energy.
(3) low amount of sarcoplasmic reticulum and more oxygen storing capacity.
(4) a reddish appearance due to high myoglobin and more ATP production via aerobic respiration.

95. Given below are two statements:

Statement I: Joints are essential for all types of movements involving the bony parts of the body.

Statement II: Force generated by the muscles is used to carry out movement through joints, where the joint acts as a load.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

96. Match excretory structure in **List-I** with animal group **List-II**.

List-I		List-II	
(A)	Protonephridia	(I)	Flatworms
(B)	Nephridia	(II)	Cockroaches
(C)	Malpighian tubules	(III)	Earthworms
(D)	Antennal glands	(IV)	Prawns

Choose the **correct** answer from the options given below:

(1) A-I, B-III, C-II, D-IV
(2) A-II, B-IV, C-III, D-I
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-I, C-II, D-IV

97. Which of the following **correctly** describes the permeability characteristics of the Loop of Henle?

(1) The ascending limb is permeable to water but impermeable to electrolytes.
(2) The descending limb is permeable to water but almost impermeable to electrolytes.
(3) Both limbs are equally permeable to water to maintain medullary osmolarity.
(4) The descending limb allows the active transport of NaCl into the medullary fluid.

98. Given below are two statements:

Statement I: The cerebral cortex contains association areas which are neither clearly sensory nor motor in function.

Statement II: The midbrain is located between the thalamus of the forebrain and the pons of the hindbrain.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

99. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: A Rh^{-ve} mother running her second pregnancy with a Rh^{+ve} foetus faces the risk of *Erythroblastosis foetalis*.

Reason R: Rh antibodies from the mother can leak into the blood of the foetus and destroy foetal RBCs.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

100. Regarding the human skeletal system, identify the **correct** statement about the ribs.

(1) All 12 pairs of ribs are connected ventrally to the sternum.

(2) The 8th, 9th, and 10th pairs of ribs are called vertebrosternal ribs.

(3) The last two pairs of ribs are not connected ventrally and are called floating ribs.

(4) Each rib is a thick cylindrical bone with a single articulation surface on its dorsal end.

■■■

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